# Jack Agterdenbos

**MSc Aerospace Engineering · Sustainable Air Transport · AI & Mathematical Optimisation**

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## PROFILE

Aerospace engineer with an MSc in Sustainable Air Transport from TU Delft (completed October 2026), specialising in AI-driven trajectory optimisation and autonomous systems. Experience of end-to-end technical delivery: built a baseline ULD availability forecasting model at KLM Cargo and sole-developed a hardware prototype for Juvoly's clinical AI pipeline. Combines a solid aerospace foundation with applied machine learning and reinforcement learning.

## SKILLS

**Programming:** Python, C++, Git

**AI & Machine Learning:** Reinforcement Learning, Goal-Conditioned RL, Supervised & Unsupervised Learning, Surrogate Modelling

**Simulation & Modelling:** Trajectory Modelling, Operations Research, Data Pipelines, Predictive Analytics

**Domain Knowledge:** Aerospace Systems, Sustainable Aviation, Airline Cargo Operations, Embedded Systems (ESP32)

## EXPERIENCE

**Intern, Cargo Control Center · KLM Cargo** · *Sep 2025 – Jan 2026*

- Built an end-to-end ULD (Unit Load Device) availability forecasting model in Python, from raw data ingestion and pipeline design to a scikit-learn model, as the baseline for later allocation work.
- Wrote the mathematical design for ULD allocation (details confidential).

**Student Prototype Developer · Juvoly** · *Jul 2025 – Present*

- Sole developer of a portable wireless audio recorder for clinical consultations, engineering the firmware in C++ on an ESP32 microcontroller with AES encryption, secure local storage, and seamless integration with Juvoly's AI transcription pipeline.
- Managed the full development cycle independently from concept to working prototype, intended for Juvoly's platform spanning 50+ medical specialisms in GP, hospital, and occupational health settings.

**Private Tutor, Maths, Physics & Chemistry · DeBijlesStudent B.V.** · *Nov 2021 – Present*

- Delivered one-to-one instruction at secondary and pre-university level.

**Earlier Roles** *Front Office Employee, Speelpark Oud Valkeveen (Jul–Aug 2022) · Grocery Delivery Driver, Picnic Technologies (Jun–Aug 2021) · Stock Clerk, Albert Heijn (May 2017–Dec 2020)*

## EDUCATION

**MSc Aerospace Engineering, Sustainable Air Transport · TU Delft** · *Sep 2024 – Oct 2026*

**Thesis: Trajectory-Based Aircraft Landing Problem (TBALP)** Formalises the TBALP under a Goal-Conditioned Hierarchical Reinforcement Learning (GCHRL) framework, implemented in Python using bluesky-gym and Stable-Baselines3. A classical planner (Constrained Position Shifting) assigns runways and arrival times, supported by an Extra Trees surrogate for fast arrival-time estimates; a Soft Actor-Critic agent adapts flight path execution to meet those targets.

**BSc Aerospace Engineering · TU Delft** · *Sep 2021 – Jul 2024*

**Semester Exchange, Business Administration · Universidad Carlos III de Madrid** · *Sep 2023 – Jan 2024*

Broadened commercial and strategic perspective alongside core aerospace training; strengthened cross-cultural collaboration in a multilingual environment.

**VWO Pre-University Education · Vituscollege** · *Sep 2019 – Jun 2021*

## LANGUAGES

**Dutch:** Native · **English:** Native / Bilingual · **German:** Limited Working Proficiency · **Spanish:** Elementary